20 cutters

Canary Cardboard Knives

WHAT'S IN THE KIT

- 20 Canary cardboard cutters

QUICK-START GUIDE

1. Demo first: saw motion, cut away from your body, flat hand never in the path.
2. Cut on a mat or scrap-cardboard surface.
3. Count knives out and count them back in — every time.

FULL LESSONS IN THIS GUIDE

PRO TIP: *These are the perfect Makedo companion — check them out together for big cardboard builds.*

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LESSON · GRADES 3-5

Precision Prototyping

LEARNING OUTCOMES

- Students will use a cutting tool with correct safe technique
- Students will improve craftsmanship in a prototype
- Students will evaluate before/after quality with criteria

MATERIALS

- Canary knives (counted out, counted back)
- Existing rough prototypes or scrap builds to upgrade
- Cutting mats or sacrificial cardboard
- Craftsmanship rubric (clean edges, straight lines, fit)

INSTRUCTIONS

1. Safety demo: saw motion, cut away from the body, flat holding hand behind the blade path.
2. Practice cuts on scrap: straight line, curve, window (interior cut).
3. Students select a rough prototype and plan upgrades: doors, windows, clean edges.
4. Upgrade session with knife-counting checkpoints.
5. Before/after photos judged against the craftsmanship rubric.
6. Reflection: how did better tools change what you attempted?

ACTIVITY

> Precision Prototyping — upgrade a rough build with clean cuts and real openings. Watching their own before/after is where maker confidence takes root.

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LESSON · GRADES 6-8

Flat-Pack Furniture Challenge

LEARNING OUTCOMES

- Students will design flat-pack pieces that assemble without fasteners
- Students will use slot-and-tab joinery principles
- Students will test assembly tolerance and refine fit

MATERIALS

- Canary knives + cutting mats
- Uniform cardboard sheets
- Rulers and pencils for measuring slots
- A real flat-pack instruction sheet as a mentor text

INSTRUCTIONS

1. Examine a real flat-pack design: how do slots, tabs, and friction replace screws?
2. Challenge: doll-scale furniture that ships flat and assembles with zero tape.
3. Teach the tolerance secret: slots must match material thickness snugly.
4. Prototype one joint first; test fit before committing to the full design.
5. Full build, then the shipping test: disassemble flat, hand to another team to assemble from a drawn instruction sheet.
6. Judge on assembly success by strangers, not on the designer's demo.

ACTIVITY

Flat-Pack Furniture Challenge — design furniture that strangers can assemble from your instruction sheet. Slot tolerance and instruction clarity fail publicly and instructively.

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LESSON · GRADES 9-12

Product Packaging Design

LEARNING OUTCOMES

- Students will reverse-engineer protective packaging design
- Students will engineer custom packaging to measurable performance specs
- Students will analyze material use vs. protection trade-offs

MATERIALS

- Canary knives + cutting mats
- Cardboard stock (limited, measured allocation)
- A 'fragile product' per team (egg, lightbulb, or ceramic tile)
- Drop-test site with standard heights; scale for weighing packages

INSTRUCTIONS

1. Autopsy session: dissect real packaging (phone box, egg carton) and name its structures.
2. Spec the brief: protect the product from a 1m drop using the least material.
3. Design and cut custom inserts: crumple zones, suspension, corner protection.
4. Weigh each package — material used is half the score.
5. Drop test at escalating heights until failure; record survival height.
6. Score = survival height ÷ package weight. Debrief the winning geometry.

ACTIVITY

Product Packaging Design — protect the egg with the least cardboard. Scoring survival-per-gram turns packaging into a genuine optimization problem, just like the industry's.